Does the galactic acceleration scale follow the cosmic age? A registered prediction, a0 ∝ 1/t, confronted with rotation curves at z = 1-2

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Version 2.42.1, 2026-10-13. Archived on Zenodo, https://doi.org/10.5281/zenodo.23329094. Licence CC BY 4.0. Developed with the help of AI assistants, as declared in the record.

Abstract

Version 2.42.1 (correction). The clock law tested here (a0 ∝ 1/t) was a translation introduced by the AI assistant, not a premise of the author, whose own reading of the distal scale is the Hubble radius c/H, i.e. a0 ∝ H(z). Section 5 of this record already found that, with a free amplitude, H describes the MUSE-DARK III bins better than the clock law (χ^2 5.1 against 14.9); with the local calibration fixed, the ordering reverses (χ^2 16.4 for the clock law and 25.2 for H). The follow-up record «Can one hidden-sector reading cover several dark-matter puzzles?» (DOI 10.5281/zenodo.23329080) adopts a0 ∝ H(z) as a phenomenological law chosen from these data and registers it for future data. CT1 stays as registered; every program, result and log is unchanged (new section 12 of the README).

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In the working records (`records/`, older logs and notes) this deposit is called 2.40, its provisional name; the environment variable MM240_SPARC_DIR keeps that name. The draft was reviewed by the second assistant before publication; the corrections are listed in `documenti/CORRECTIONS_before_publication.md`.

Version 2.42.1 (October 2026) adds section 12, a correction: the clock law tested here was a translation introduced by the AI assistant, not the author's premise; the follow-up record adopts a0 ∝ H(z), chosen from the MUSE-DARK III bins, and registers it for future data. Every program, result and log is unchanged.

MM is the author's framework. This record tests the assistant's cosmic-age parametrisation of the author's distal-scale idea; section 12 (version 2.42.1) clarifies that distinction.

In brief

- The radial acceleration relation of disc galaxies has one scale, a0 ≈ 1.2 × 10^{-10} m s^{-2}, numerically close to cH0/2π (Milgrom 1983). Whether a0 changes with cosmic time can be tested with rotation curves at z ~ 1-2 (Milgrom 2015, 2017). MUSE-DARK III (Ciocan et al. 2026) reports a growth of a0 with redshift.
- This record fixes one law before these comparisons (the data sets were already known). **CT1: a0(z)/a0(0) = t0/t(z)**, with t the cosmic age computed from the expansion history of MM 2.32 (all 8896 samples of its chain) and the clock started at the beginning of the model. It predicts a0 larger by a factor 2.35 at z = 1 and 4.18 at z = 2, with a 95 % cosmological band of ±0.42 % at z = 1 and ±0.49 % at z = 2. In the independent check of section 1 the same clock in ΛCDM differs by 0.14 % at z = 1 and 0.35 % at z = 2 (compared sample by sample, by up to 1.8 % at z = 2.5): the test is a test of the clock, not of the MM cosmology.
- Confronted with four published data sets. The first three prefer a constant a0 at face value:
- **RC100** (100 galaxies, z = 0.6-2.5): at z ≥ 2 the dark-matter fraction at the effective radius predicted from the kinematics is 0.24 with a constant a0 and 0.63 with CT1; the published value is 0.26.
- **MUSE-DARK II** (95 low-mass galaxies at z ≈ 1): the zero point of the baryonic Tully-Fisher relation does not evolve (0.00 ± 0.06 dex); in the low-acceleration regime assumed here CT1 predicts −0.26 to −0.32 dex.
- **KMOS3D** (massive galaxies): the zero point rises by 0.17 dex from z ≈ 0.9 to z ≈ 2.3; CT1 predicts a fall of about 0.2 dex.
- **MUSE-DARK III** (four bins at z = 0.5-1.3, from a digitization of the published figure) goes the other way: a0 grows with redshift, and a constant a0 is the worst law. With the local calibration frozen, CT1 has a better nominal statistic than a0 ∝ H; with a free amplitude, H describes the shape better. The growth is slower than CT1's: 1.36 between the last and the first bin, against 1.75. The data sets are in tension with each other; their different mass normalisations are a possible explanation, not one isolated by this record.
- For RC100 at z ≥ 2, the mass shift that minimises the nominal χ^2 of CT1 corresponds to about half the adopted baryonic mass (0.78 times at z < 1.2); in the regime assumed, MUSE-DARK II needs a similar factor (0.48-0.55). At z ≥ 2, at the central values, the mass CT1 requires is below the stellar mass for 26 RC100 galaxies out

of 41. Gas measurements in samples of star-forming galaxies at z ≈ 1-2 point the other way: molecular gas from CO lines and dust, calibrated on 1444 galaxies with three methods, is comparable to or larger than the stars at z ≈ 2, and atomic hydrogen at z ≈ 1 has on average the stellar mass. They are population results, not measurements of the galaxies compared here. **CT1, as registered, is in tension with the comparisons adopted; the tension is conditional on the mass models and on the simplifications declared.**

1. The law tested (CT1)

Statement. a0(z) = a0(0) · t0/t(z), with t(z) = ∫_z^∞ dz′/[(1 + z′) H(z′)] the proper age of the comoving FLRW congruence, H from MM 2.32.

Assumptions (they are the bridge; they are not derived from the 2.32 equations):

1. the clock of the medium, U^μ∇_μT = 1 with T = 0 at the initial surface of the model, is the cosmic age: T = t, with no formation age or offset per galaxy;
2. a0 = ξc/T with ξ universal and constant; ξ is calibrated from the local value a0(0) = 1.2 × 10^{-10} m s^{-2}: ξ = 0.174 (a calibration, not a prediction);
3. the same a0(z) enters the radial law already adopted in the programme (the "MLS" form of the radial acceleration relation), g_obs = g_bar/[1 − exp(−√(g_bar/a0))];
4. the galaxies tested are in the regime of validity of that law; external fields, pressure and selection are treated in the same way for every hypothesis.

Numbers (`codice/ct1_previsione.py`; 95 % band of the cosmological propagation only):

z	CT1, a0(z)/a0(0)	same clock in ΛCDM (independent check)	a0 ∝ H	baryonic Tully-Fisher shift, asymptotic (dex)
0.5	1.6025 [1.5975, 1.6081]	1.6024	1.325	−0.205
1.0	2.3476 [2.3378, 2.3574]	2.3508	1.787	−0.371
1.5	3.2127 [3.1978, 3.2274]	3.2211	2.356	−0.507
2.0	4.1810 [4.1606, 4.2014]	4.1956	3.009	−0.621
2.5	5.2409 [5.2143, 5.2676]	5.2622	3.734	−0.719

Present age t0 = 13.778 Gyr [13.744, 13.814]. The independent check uses a flat ΛCDM background with the chain medians (Ωm = 0.3066, h = 0.6764) and no MM code.

Alternatives on the same background and normalisation: a constant a0; a0 ∝ H, considered by Milgrom (2015). In Verlinde (2016) the cosmological acceleration cH0 sets the scale and the MOND-like scale is a_M = cH0/6; that work does not derive the replacement H0 → H(z), and it discusses the cosmological limits of its approximation.

What a success would and would not show (`documenti/SHAPE_TEST_REPORT_astra_IT.md`). The law makes 1/a0 affine in the look-back time. Three degeneracies are exact: a constant a0 satisfies the affine form with zero slope; with a free clock zero the ratio tends to one and imitates a constant; in the matter era 1/t and H are proportional. With the clock zero fixed, CT1 has no freedom left; a success would support "a0 follows the cosmic age", not the MM cosmology in particular. In the design study of the second assistant (five bins, the synthetic scheme of the pilot), separating CT1 from a two-parameter alternative such as a0 ∝ E^γ needed a precision of about 3.4 % per bin in ln a0: a result of that scheme, not a general requirement.

2. Test 1: dark-matter fractions at the effective radius (RC100)

Data. Table 3 of Nestor Shachar et al. 2023: for each galaxy Re, Vc(Re) and the dark-matter fraction fDM(Re) from the authors' dynamical decomposition (`dati/rc100_table3.csv`). In that decomposition the baryonic mass is a free parameter of the fit, with a Gaussian prior centred on the stellar mass from SED fitting plus the gas mass from the

scaling relations of Tacconi et al. (2020); the table lists the fitted value. One galaxy (U3 25105, z = 2.52) lies just beyond the grid of the CT1 samples (z ≤ 2.5) and is extrapolated in tests 1 and 3: the extrapolated step from z = 2.50 to 2.52 (a factor 1.00829) differs by 0.016 % from the exact step of the same clock in ΛCDM (1.00845); that difference mixes the MM-ΛCDM offset and the extrapolation. Applied to the exact ΛCDM curve on the same grid, the same extrapolation errs by −0.014 % for the clock and −0.007 % for H (both printed by `codice/controllo_z2.py`).

Method (`codice/controllo_z2.py`, protocol `records/PROTOCOLLO_CONTROLLO_Z2.md`). From the kinematics, g_obs = Vc^2/Re. For each hypothesis the MLS law is solved for g_bar and the predicted fraction f_pred = 1 − g_bar/g_obs is compared with the published fDM. The uncertainty combines the error on fDM and the errors on Vc and Re propagated through f_pred; errors are treated as independent. A common shift of log g_bar (a systematic error of the baryonic masses) is scanned from −0.6 to +0.3 dex; the shift reported minimises the nominal χ^2, and by itself does not show a full reconciliation.

41 galaxies at z ≥ 2 (median z 2.22)	predicted fraction (median)	published (median)	nominal χ^2	below −2σ	mass shift minimising χ^2
a0 constant	0.24	0.26	68.9	2	0.00 dex
a0 ∝ H	0.55	0.26	189.6	17	−0.21 dex
CT1	0.63	0.26	278.5	19	−0.32 dex

At z < 1.2 (32 galaxies): χ^2 68.2 with a constant a0 and 101.8 with CT1, which would need −0.11 dex. The four registered expectations (E1-E4) all held. χ^2 per galaxy is 1.7-2.1 even for a constant a0: the nominal errors may be narrow, or neglected covariances and imperfections of the model contribute. This goes in the direction of Milgrom (2017), who wrote from six galaxies of Genzel et al. (2017) that these data "all but exclude" a0 ≈ 4a0(0) at z ~ 2.

3. Test 2: the baryonic Tully-Fisher relation (MUSE-DARK II, KMOS3D)

Why not RC100 again. A Tully-Fisher relation built from the same galaxies would repeat test 1.

Method (`codice/controllo_tully_fisher.py`, protocol `records/PROTOCOLLO_CONTROLLO_TF.md`). With the MLS law, at fixed radius and velocity, a galaxy in the regime y0 = g_bar/a0(0) needs, under a0(z) = R(z)a0(0), a baryonic mass k times the one needed with a0(0): the zero-point shift is log10 k. In the deep regime it tends to −log10 R(z); in the Newtonian regime to zero. Sizes are held fixed.

MUSE-DARK II (independent of RC100). The galaxies are small and slow. This record assumes that their velocities at 2Re lie at low accelerations (y0 ≲ 0.3), where the relation depends on a0 more than on sizes; the shifts below hold for that regime. The representative galaxy of the source ($M^* = 10^9$ solar masses, rotation velocity 70 km/s at 1.8Re with dispersion 30 km/s, hence a circular velocity of about 86 km/s) does not by itself establish the regime, because its radius and baryon distribution are not given.

hypothesis	predicted shift (y0 0.05-0.3)	distance from 0.00 ± 0.06 (published fit)	adding a common ±0.20 dex (a scenario chosen here)
a0 constant	0.00	0 σ	0 σ
a0 ∝ H	−0.17 to −0.21	2.8-3.6 σ	0.8-1.0 σ
CT1	−0.26 to −0.32	4.3-5.3 σ	1.2-1.5 σ

The ±0.06 dex of the published fit already includes the ±0.20 dex that the authors assign to the individual baryonic masses; the last column adds the same amount again as an error common to all masses, a scenario of this record, not a propagation prescribed by the source. To reconcile CT1, the true baryonic mass would be 0.48-0.55 times the estimate: with the stars fixed, only 7-20 % of the estimated gas would remain. The gas share of the estimate, 56 %, is an assumption for a representative galaxy: the source gives separately the medians of molecular (34 %) and atomic gas (22 %), and the sum of two medians need not be the median of the sum.

KMOS3D (same population as RC100, not independent). Measured: the zero point rises by +0.17 ± 0.06 dex from z ≈ 0.9 to z ≈ 2.3. At fixed regime CT1 predicts a fall of 0.17-0.21 dex, a0 ∝ H of 0.13-0.17 dex, a constant a0 no change. The fixed-regime model does not reproduce the rise for any law; a structural evolution (more baryons within the radius

of the velocity) is one possible explanation.

Registered expectations: T1 did not hold (in the assumed regime CT1 gives −0.26 to −0.32, not −0.30 to −0.37), T2-T4 held. The protocol contains an arithmetic error (a representative velocity of 61 km/s instead of 77.8 km/s), declared in `records/NOTA_ESITO_CONTROLLO_TF.md`; the regime remains an assumption, as stated above.

4. Test 3: are the masses overestimated? Stars and directly measured gas

Method (`codice/controllo_gas.py`, protocol `records/PROTOCOLLO_CONTROLLO_GAS.md`). For each RC100 galaxy the baryonic mass that a hypothesis requires, at fixed radius and mass distribution, is Mbar_req = Mbar · (1 − f_pred)/(1 − fDM). It is compared with the stellar mass from SED fitting, which does not depend on gas. Test 3 reuses the RC100 data and the predicted fractions of test 1. The gas scaling relation adopted (Tacconi et al. 2018, Table 3b) is a reduced version: it omits the size term 0.11 log(Re/Re,0).

41 galaxies at z ≥ 2	required baryonic mass minus stellar mass (median)	galaxies below their stellar mass	implied gas fraction (median)
a0 constant	+0.17 dex	7	33 %
a0 ∝ H	−0.04 dex	23	negative
CT1	−0.16 dex	26	negative

At z < 1.2 CT1 requires +0.11 dex above the stars (implied gas 23 %), a constant a0 +0.22 dex (39 %). The scaling relations give, at z ≥ 2, a gas mass 1.32 times the stars in the median; the dynamical masses of RC100 already lie 0.19 dex below stars plus scaling-relation gas, and 0.22 dex above the stars. These comparisons use central values and fixed mass distributions; the stellar and dynamical uncertainties are not propagated.

The gas measured directly. The scaling relations of molecular gas are fits to 1444 star-forming galaxies at z = 0-4 measured with three independent methods (CO lines, far-infrared dust emission, dust photometry at about 1 mm), which follow the same trends once their zero points are aligned (Tacconi et al. 2018); the ratio of molecular gas to stars grows as (1 + z)^2.5. Atomic hydrogen at z ≈ 1, measured by stacking the 21-cm signals of star-forming galaxies, has on average the same mass as their stars (Chowdhury et al. 2020); an average from stacking is not the median fraction within the kinematic radius. In these populations, then, gas is a large share of the baryons (about 35-60 %, depending on the population and the gas component). These are not measurements of the galaxies compared here. In the regime assumed, CT1 allows gas to be 8-21 % of the baryons of the representative MUSE-DARK II galaxy (a different quantity from the 7-20 % of the estimated gas above) and, at the central values, a negative amount in the median for RC100 at z ≈ 2. A common error of the zero points (CO-to-H2 conversion factor, dust-to-gas ratio) of 0.1-0.2 dex is possible; under these assumptions it would not be enough. The four registered expectations (G1-G4) all held.

5. Test 4: the growth reported by MUSE-DARK III (z = 0.5-1.3)

Data. Four bins of a0(z) (`dati/muse_dark3_bins_alcock.csv`): Table 1 of Alcock (MNRAS, DOI 10.1093/mnras/stag1485), a digitization of Fig. 3 of Ciocan et al. (MUSE-DARK III). They summarise a dynamical analysis of the galaxies; they are not new measurements. The table gives neither the confidence convention nor the covariance of the bins; the bars are symmetrized.

Method (`codice/controllo_muse_dark3.py`, protocol `records/PROTOCOL_CLOCK_TEST_astra.md`, written before the calculation but after the published data were known). a0(0) is calibrated on SPARC alone (149 galaxies, 3150 points; MLS law; mass-to-light ratios 0.5 for discs and 0.7 for bulges; equal weight per galaxy) and then frozen: a0(0) = 1.019×10^{-10} m s^{-2} (bootstrap 95 %: 0.90-1.16; 1.29 and 0.83 with the stellar masses, discs and bulges, × 0.8 and × 1.2, the gas being fixed). Each law predicts the four bins with no fit on the distant data. A second diagnostic frees one amplitude per law and tests the shape alone. The program reads the bars of the bins as 1σ with a diagonal observational covariance (this gives the χ^2 below). The further scenarios of the protocol (bars read as 95 %, correlations between bins, extra systematics) are reported in `documenti/CLOCK_TEST_REPORT_astra_IT.md` and are not regenerated by the code distributed here.

law	predicted a0 at z = 0.50 / 0.83 / 1.05 / 1.28	nominal statistic (4 bins)	shape χ^2 (3 degrees of freedom)	last bin / first bin
data	1.99 / 2.20 / 2.57 / 2.71	—	—	1.36

a0 constant	1.02 / 1.02 / 1.02 / 1.02	168.6	28.8	1.00
a0 ∝ H	1.35 / 1.64 / 1.87 / 2.13	25.2	5.1	1.57
CT1	1.64 / 2.11 / 2.47 / 2.86	16.4	14.9	1.75
1 + z (chosen after seeing the data)	1.53 / 1.86 / 2.09 / 2.32	11.8	3.3	1.51

Units 10^{-10} m s^{-2}. The nominal statistic includes the errors of the bins, the spread of the local calibration and the cosmological spread of the samples. The same clock in ΛCDM differs from CT1 by at most 0.2 % at these redshifts. Tests 1-3 use $a0(0) = 1.2 \times 10^{-10}$ m s^{-2}; test 4 calibrates its own value, 1.019×10^{-10} m s^{-2}, on SPARC. The shape test does not depend on this difference.

Result. The MUSE-DARK III bins favour an evolution of a0 over a constant, which is the worst law. With the local calibration frozen, CT1 has a better nominal statistic than a0 ∝ H (16.4 against 25.2); with a free amplitude, H describes the shape better (χ^2 5.1 against 14.9 for 3 degrees of freedom). The central growth of CT1 is steeper than that of the bins: CT1 is too low at z = 0.5 and too high at z = 1.3. This comparison does not give the robust support that was required, but it goes in a different direction from the first three. The law 1 + z fits best, but it was chosen after seeing the data and is only a comparison. The test was run by the second assistant and recomputed by Claude with a different code, which is the one in `codice/`; the numbers agree (`records/NOTA_ESITO_CONTROLLO_MUSE_DARK3.md`).

6. What the tests say together

- In the first three comparisons CT1 needs, at face value, baryonic masses of distant galaxies smaller than those used: about half for RC100 at z ≥ 2 and for MUSE-DARK II in the regime assumed, 0.78 times for RC100 at z < 1.2 (shifts that minimise the nominal statistics); at z ≥ 2, at the central values, fewer baryons than stars for 26 galaxies out of 41. A constant a0 describes RC100 and the simplified MUSE-DARK II comparison better than CT1. The fixed-regime model does not reproduce the rise of the KMOS3D zero point (+0.17 ± 0.06 dex) for any law; a structural evolution is one possible explanation.
- The two MUSE-DARK papers differ in the mass normalisation, as well as in sample, selection and observable: paper III (dynamical disc normalisation) finds a growth of a0; paper II (stellar masses from SED fitting and gas from scaling relations) finds no evolution of the baryonic Tully-Fisher relation. The mass normalisation is a possible explanation of the difference, documented in the papers; this record does not isolate it.
- Test 4 shows the tension from the other side: with the MUSE-DARK III bins a0 grows, but more slowly than CT1. A constant a0 at z ≈ 2 (RC100) and a growth at z = 0.5-1.3 (MUSE-DARK III) are in tension; they are not logically incompatible (a non-monotonic evolution, different populations or systematics could reconcile them). None of the four comparisons gives CT1 the robust support required in its registered form.
- Gas measurements in populations of star-forming galaxies confirm large gas fractions at z ≈ 1-2. They do not measure the galaxies compared here, and the stellar and dynamical uncertainties are not propagated: "the masses are overestimated" is therefore not excluded, but it would have to hold against these population results.

7. What is not claimed

- That a0 is constant: the comparison is conditional on published masses and decompositions.
- That CT1 is formally excluded: the comparisons are conditional on published mass models and no joint likelihood with covariances has been built. In its registered form, at the central values of the RC100 masses, it would need fewer baryons than stars at z ≥ 2 for 26 galaxies out of 41.
- That MM is tested: only the clock law is tested; the MM expansion history changes the central prediction by 0.35 % at z = 2 (independent check of section 1).
- That the MUSE-DARK III bins are independent measurements: they are a digitization of a published figure, and test 4 was written after they were known; the law 1 + z was chosen after seeing them.
- That the tests are independent of each other: KMOS3D and RC100 share the population, and test 3 reuses test 1; tests 1-3 depend on gas scaling relations (the RC100 priors, the MUSE-DARK II gas); test 4 depends on the SPARC calibration and on the dynamical normalisation of MUSE-DARK III.
- That a0(0) or ξ is derived: both are calibrated. No galaxy data modify R(z); the normalisation is calibrated locally and the fits on distant data are diagnostic only.
- That a different clock would fare better: a clock started later would be a new hypothesis, to be written down before independent data, not adjusted on these.

8. How to reproduce (Python 3 with numpy, scipy, matplotlib)

```
python codice/run_all.py        # the five programs; outputs in uscite/, compared with risultati/
(about one minute);
                                # exit code 0 only if every log and JSON is reproduced identically
python codice/ct1_previsione.py
python codice/controllo_z2.py
python codice/controllo_tully_fisher.py
python codice/controllo_gas.py
python codice/controllo_muse_dark3.py
```

Test 4 needs two files of the SPARC database (Lelli, McGaugh and Schombert 2016), not redistributed here: `SPARC_Lelli2016c.mrt` and `Rotmod_LTG.zip` from http://astroweb.case.edu/SPARC/, to be put in `dati/esterni/` (or in the folder named by the environment variable MM240_SPARC_DIR). The program checks their SHA-256 fingerprints, listed in `dati/README.md`.

`documenti/calcola_ct1_astra.py` is the program that produced `dati/ct1_campioni.npz`, included as a historical source: it does not run as distributed. It needs the MM 2.32 reproduction kit (Zenodo, DOI 10.5281/zenodo.22840735) and CAMB 2.0.4, its paths are those of the author's machine, and it reads its protocol as `PROTOCOLLO_CT1.md` next to itself (distributed here as `records/PROTOCOL_CT1_astra.md`): without that file it writes the samples and then stops while building its summary. Code comments and printed labels are partly in Italian. SHA-256 fingerprints are in `IMPRONTE.sha256`.

9. Provenance

The author is not a professional physicist. The framework and its premises are his. The computations were done with AI assistants: the CT1 samples, the shape test and test 4 by the second assistant (Codex, "Astra"), the first three tests by Claude; test 4 was then recomputed by Claude with a different code, which is the one in `codice/`. Each protocol was written and hashed before its calculation (`records/`); the hashes are not publicly time-stamped. One registered expectation (T1) did not hold and one protocol contains an arithmetic error: both are reported. Before publication the second assistant reviewed the draft; its reports (two rounds) are in `documenti/REVIEW_astra_20261011_IT.md` and `documenti/REVIEW_astra_20261011_round2_IT.md`, and the corrections are listed in `documenti/CORRECTIONS_before_publication.md`. The reference outputs were then regenerated with Python 3.13.13, numpy 2.4.6, scipy 1.18.0, matplotlib 3.11.1: the numbers are unchanged, some printed labels are corrected. A sentence of the earlier working notes on the origin of the RC100 masses was inexact and is corrected in `records/NOTA_ESITO_CONTROLLO_GAS.md`. No human referee has checked this record.

10. Glossary

term	meaning here
a0	the acceleration scale of the radial acceleration relation
MLS law	g_obs = g_bar/[1 − exp(−√(g_bar/a0))], the form of the radial acceleration relation used by the programme
Re	effective (half-light) radius
fDM(Re)	the fraction of the circular speed squared at Re not accounted for by baryons
zero point	the intercept of log Mbar against log V at fixed slope
dex	a factor of ten in logarithm: 0.3 dex is a factor two
CT1	the law a0 $\propto$ 1/t with the clock started at the beginning of the model

11. References

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- G. T. Alcock, Is the acceleration scale of the radial acceleration relation tracking the cosmic expansion rate?, MNRAS 551 (2026), DOI 10.1093/mnras/stag1485, arXiv:2609.38796 (Table 1: binned values of a0(z) read from Fig. 3 of MUSE-DARK III).
- U. Lanciano, Mathematical Materiality 2.32, Zenodo, DOI 10.5281/zenodo.22839525 (2026).

12. Correction (added in version 2.42.1, October 2026)

- **What changes.** The identification of the acceleration scale with a clock that reads the cosmic age (assumptions 1-2 of section 1) was a translation introduced by the AI assistant. It is not a premise of the author. The author's own reading of the distal scale is the Hubble radius R_H. With a0 proportional to c^2/R_H and a redshift-independent proportionality factor, identifying $R_H = c/H$ gives $a0 \propto H(z)$, not $a0 \propto 1/t$.
- **What the data already said here.** Section 5 reports that, with a free amplitude, the H law describes the MUSE-DARK III bins better than CT1 (χ^2 5.1 against 14.9 for 3 degrees of freedom). With the local calibration frozen, CT1 was nominally better (16.4 against 25.2).
- **The follow-up.** The follow-up record, *Can one hidden-sector reading cover several dark-matter puzzles?* (Zenodo, October 2026; see the related works of this record), repeats the comparison with $\Omega m = 0.315$ (χ^2 5.5 for H against 15.3 for the age law). It adopts $a0(z) = A \cdot H(z)/H0$ as a phenomenological law chosen from these data, not as a confirmed one. It registers this law for future data, with the shape tested separately from the normalisation (A free for each homogeneous sample, the local value treated as a separate test).
- **What does not change.** CT1 stays as registered here, with its four comparisons and the tensions of section 6. Every program, result and log of this record is identical to version 2.42.

Sintesi in italiano

- **La domanda.** La scala di accelerazione delle galassie, a0, cambia con l'età dell'universo? Si può controllare con le curve di rotazione delle galassie lontane, a z fra 1 e 2.
- **La legge provata (CT1), fissata prima di questi calcoli, con dati già conosciuti.** a0 è inversamente proporzionale all'età dell'universo, calcolata con la cosmologia della MM 2.32: a z = 1 vale 2,35 volte quella di oggi, a z = 2 vale 4,18 volte. Con la cosmologia standard la curva è quasi identica (0,35 % di differenza a z = 2 nel controllo indipendente): si prova l'orologio, non la cosmologia MM.
- **Tre confronti con dati pubblicati.** Le galassie di RC100 a $z \geq 2$ hanno la materia oscura apparente che dà un a0 costante (26 % misurata, 24 % prevista), non quella di CT1 (63 %). Le galassie piccole di MUSE-DARK II a $z \approx 1$ seguono la stessa Tully-Fisher barionica di oggi; nel regime di bassa accelerazione assunto qui, CT1 prevede che a parità di velocità abbiano circa la metà della massa. Le galassie grandi di KMOS3D vanno nel verso opposto a CT1.

- **Il quarto confronto (MUSE-DARK III, z 0,5-1,3, da una digitalizzazione della figura pubblicata).** Qui a0 cresce, e a0 costante è la legge peggiore; con la calibrazione locale fissata CT1 fa meglio di $a0 \propto H$ (16,4 contro 25,2), ma con ampiezza libera H descrive meglio la forma (χ^2 5,1 contro 14,9). a0 cresce più piano di CT1: 1,36 fra l'ultimo e il primo bin, contro 1,75. I dati sono in tensione fra loro; una spiegazione possibile sono le diverse normalizzazioni delle masse, che il deposito non isola.
- **Che cosa servirebbe a CT1.** Masse barioniche delle galassie lontane minori di quelle usate (circa la metà a $z \geq 2$): a $z \geq 2$, ai valori centrali, meno barioni delle sole stelle per 26 galassie su 41. Il gas misurato in popolazioni di galassie simili (CO e polvere su 1444 galassie con tre metodi; idrogeno atomico a $z \approx 1$ in media pari alle stelle) va nel verso opposto, ma non misura le galassie confrontate qui. Così com'è registrata, CT1 è in tensione con questi confronti; la tensione dipende dai modelli di massa adottati.
- **Onestà.** Protocolli scritti prima dei conti; un'attesa registrata (T1) non si è verificata e un protocollo contiene un errore di conto: tutti e due sono dichiarati. Prima della pubblicazione il testo è stato rivisto dal secondo assistente; le correzioni sono elencate in `documenti/CORRECTIONS_before_publication.md`. L'autore non è un fisico di professione; i calcoli sono fatti con assistenti IA.
- **Correzione (versione 2.42.1).** L'orologio che legge l'età cosmica è stato una traduzione dell'assistente, non una premessa dell'autore. La lettura dell'autore della scala distale è il raggio di Hubble c/H, che dà $a0 \propto H(z)$. Già qui, con ampiezza libera, H descrive meglio i bin di MUSE-DARK III. Il deposito successivo (vedi i collegamenti del record) adotta $a0 \propto H(z)$ come legge scelta dai dati e la registra per i dati futuri. CT1 resta come registrata; programmi, risultati e log sono invariati.